XV.

QVOMODO IMPERATOR REGNI GVIANÆ, NOBILES SVOS ORnare & præparare ſoleat, ſi quando ad prandium vel cœnam eos inuitare velit.

INcolæ regni GVIANÆ, quemadmodum etiam vicini populi, ebrietati admodum dediti ſunt, & omnes nationes potando ſuperant. Quando Imperator ſuis nobilibus & regni proceribus prandium facere cupit, omnes ad prandium inuitandi, depoſitis veſtimentis, nobiliſſimo balſamo albi coloris, à vertice vſque ad calcem inunguntur, & poſtea auro in puluerem tenuiſſimum redacto aſperguntur, vt toti aurei appareant. Hoc facto, quinquageni & centeni ſimul accumbunt, totis ſeptem vel octo diebus continuè potantes, donec amplius non poſſint. Cum Angli aliquando ad CASSIQVE TOPARIMACA veniſſent, qui aliũ quendam CASSIQVE inuitauerat, inuenerunt eos in BRESILIENSI HA-

Shimmering Worlds: Brilliance, Power, and Gold in Pre-Columbian Panama

NICHOLAS J. SAUNDERS

The history of the Americas shimmers with gold. Since Columbus's fateful encounter in 1492, that incorruptible metal has symbolized the wealth of a continent and forged an enduring fascination with the pre-Columbian past. Yet the European obsession with gold has often obscured more than it revealed of those who created extraordinary works of art from the metal and its alloys. The indigenous worlds of prehistoric American peoples—their ways of perceiving and valuing the world—were reflected in the shiny objects they made. But the native realities and ancient systems that sustained their universe were subsumed beneath the Europeans' all-consuming lust for gold as a material object. Modern archaeology too has often been seduced by the lure of gold as a metal. Scientific analyses of ancient metalworking traditions have often focused on technology and ignored indigenous notions of power and the ways in which spirituality and identity were fused in what has been called the "aesthetic of brilliance" (Saunders 1998; 2004). The many legends of El Dorado bear witness to western civilization's obsession with its own notions of reality rather than with the actual lives of ancient societies (Mason 1990; Whitehead 1997: 72–75).

Facing: Origins of the myth of El Dorado, The Gilded Man. An Amerindian chief has gold dust blown over his resin-covered body. *The American Indians Depicted in a Collection of Plates Engraved by Theodore de Bry in the Years 1590–1602.* GM 2975.1585

Chief Francisco Pronhōpa of the Xavante Tribe.

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Today, as in the recent past, indigenous Americans tend to see the world as a place where an object's physical and spiritual qualities integrate with and are vivified by phenomenological experience. This philosophy, so dramatically different from 16th-century European ideas, and modern western ones as well, was likely shared to a considerable degree by pre-Columbian civilizations.

It can be argued that all the indigenous peoples of the Americas saw shiny objects, places, and phenomena as containing elements of the spiritual and creative power of light. Brilliant things—both cultural and natural—were regarded as effervescent concretizations of light. Thus, golden jewelry, feather headdresses, lightning and sunlight, were all charged with cosmological power.

In Panama, as elsewhere in the Americas, gold and other kinds of shiny things existed not in isolation but within the overlapping spheres of symbolic, ritual, and technological activity. Panama's pre-Columbian golden treasures, therefore, should be seen from a broad perspective, where meanings are attached to objects, and the objects themselves are embodiments of native philosophies and worldviews. Golden artifacts are statements of "social being" which define and manifest the relationships between humans and supernatural forces.

SENSUAL WORLDS

A wealth of ethnohistorical and ethnographic evidence suggests that indigenous American peoples saw (and still see) the world as infused with "spiritual brilliance." The sun, moon, water, ice, rainbows, and clouds glisten with this shiny essence, as do minerals, feathers, animal pelts, pearls, and shells. The artifacts of culture, too, share this glittering quality, from ceramics to textiles and metals (Saunders 1998: 226–230). Sacred ritual knowledge can also be referred to as shining and brilliant—hardly surprising given that rituals (and the knowledge required to conduct them) are the social mechanisms by which a society creates and maintains itself; shiny objects and brilliant knowledge simultaneously validate and reinforce each other as the efficacious possessions of high status individuals, whether shamans, priests, chiefs, or kings.

From the Amazon to the Andes, from Lower Central America through Mesoamerica and the Caribbean to North America, various cultural traditions interpret these ideas in their own ways. Particular beliefs and symbolic associations, and specific technological ways of doing things, are nevertheless underwritten by a pan-Amerindian appreciation of brilliant things. Burnished pottery, shimmering featherwork, polished wood, gemstones, obsidian, and a variety of metals all manifest shininess but are chosen and used differently by individual societies. Each of these items is embedded in the mythology, religion, ideology, and everyday reality of individual societies in distinctive ways.

Different societies gave their own characteristic values and meanings to the shapes, colors, and textures of their shiny objects (Hamell 1995: 47–49; Melgarjo 1998; Reichel-Dolmatoff 1978). Yet, despite these differences, attitudes toward brilliant objects emerged from and reinforced a worldview that regarded light, color, and shiny matter as indicating the presence of supernatural beings and essence in the earthly world. Such potent objects required equally powerful individuals to control them through special knowledge that was the preserve of the local elite.

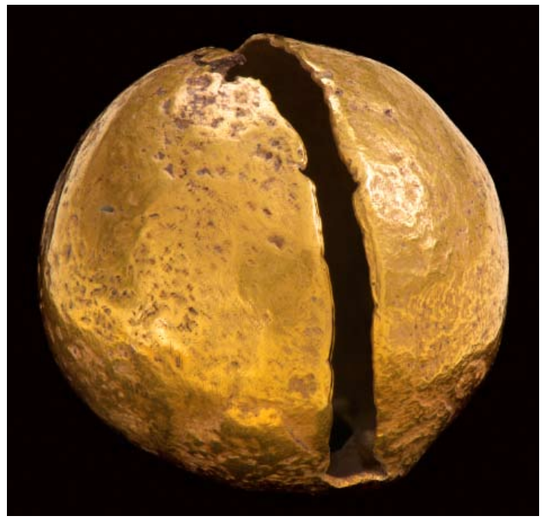
For indigenous American peoples, smell, sound, and touch joined with vision to create a multisensory world of phenomenological unity. In this holistic universe, what Europeans regarded as illogical meanings were attached to the sounds, sights, smells, and tastes of life and death (Howes 1991: 3–5; Classen 1990, 1991). The odors of cooking, sex, and putrefaction are sensed by the nose; the sounds of speech and music (especially perhaps the sound of gold/copper bells) and the natural world are heard by the ears. The nose and ears are sensory organs that are adorned with mineral and/or golden artifacts, perhaps to render safe or civilized such sensory stimuli they receive. In this way, invisible and ephemeral sensations and esoteric knowledge are rendered material by leaving an artifactual trace in the archaeological record.

The natural world was a resource from which cultures fashioned a series of interleaving and overlapping cultural realms and bestowed significance on fauna, flora, and natural phenomena—often recombining elements from each to signify a culture-specific meaning or set of meanings. Amerindians, like the members of all human societies, created their own ethno-taxonomies of the world, and each mineral, plant, and animal had its own place and significance in the semantic framework of life. This framework was a matrix composed of categories of being that were formed partly by the observation of reality and partly by metaphorical attributes and connections specific to a society's own myths and beliefs. Some animals, therefore, were selected to represent particular human qualities, and in ancient Panama, crocodiles, bats, and the hook-beaked bird seem to have represented warrior qualities and appear in gold (Cooke et al 2003:137). No animal symbol existed by chance, and no golden artifact in animal shape was devoid of a set of specific and probably evocative sensorial meanings.

Bat effigy pendant, Gran Coclé, 500–1520, gold. GM 5645.226

Nose ornament, Gran Coclé, 500–1520, gold. GM 5645.13

Ear spool, Gran Coclé, 700–1000, gold. GM 5645.12

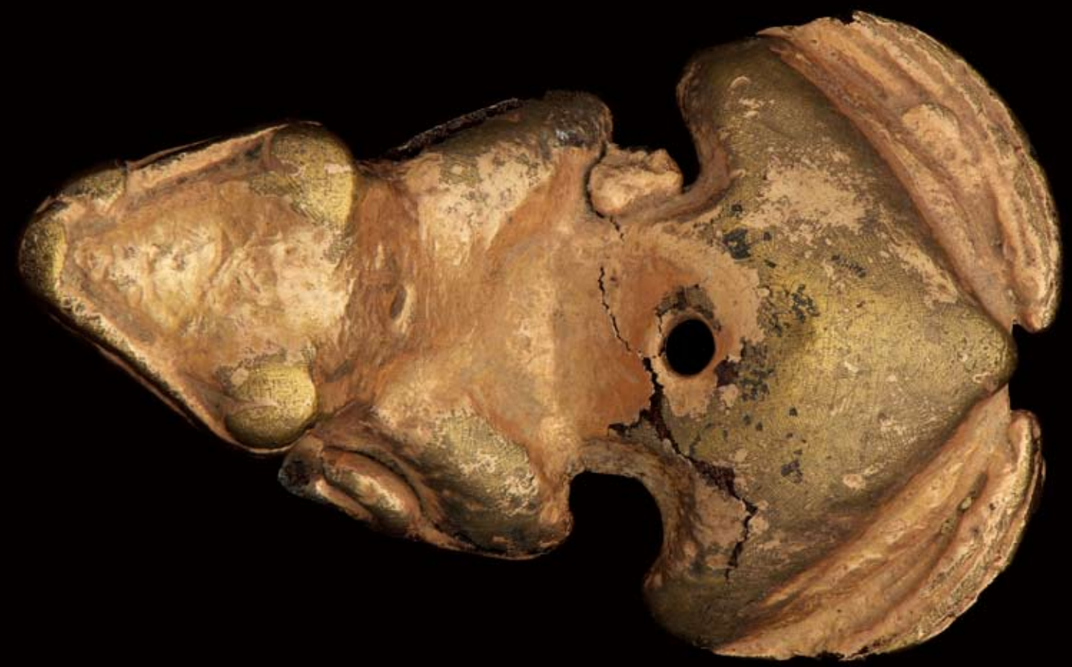


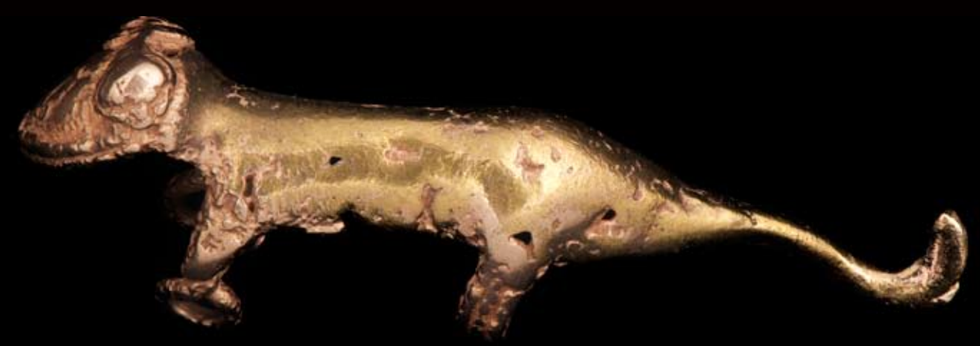
Above: Clockwise: Bell, Gran Coclé, 700–1000, gold. GM 5645.129; Ear spool, Gran Coclé, 700–1000, gold. GM 5645.88; Nose ornament, Gran Coclé, 700–1000, gold. GM 5645.84; Nose ornament, Gran Coclé, 700–1000, gold. GM 5645.130

Facing: Ear rod finial, Gran Coclé, 700–1000, gold. GM 5645.167

It is by no means clear or obvious that a "realistic" zoomorphic shape was meant to represent the animal per se. Various animals can be shamans' spirit helpers—sometimes also the transformed shaman himself—or malevolent ancestors. An animal image is a bundle of culturally specific meanings and qualities rather than a simple depiction of a particular creature. This is one reason why some golden images combine parts of different animals, either recognizably or in fantastical shapes, and sometimes add zoomorphic elements to otherwise obviously human forms.

Birds are well known shamanic familiars and a rich source of sensual symbolism in current ethnographic societies of South America, where they are related to tobacco curing and creation myths (Reina and Kensinger 1991; Wilbert 1985). The flapping of a bird's wings, its distinctive song, the color and shape of its beak, and the dazzle of its iridescent feathers were all natural features reinterpreted and re-presented in mythology and materialized in zoomorphic artifacts such as the golden bird images and polychrome bird-shaped whistles in the Gilcrease Collection (see Cooke, this volume, page 138).





Frog effigy pendant, Gran Coclé, 500–1520, gold. GM 5645.96

Composite effigy pendant, Gran Coclé, 500–1520, gold. GM 5645.15

Double winged pendant, Gran Coclé, 500–1520, gold. GM 5645.99

Endangered
Panamanian golden frog
on a branch.
©Gary Scott

Similarly, the silvery glisten of fish scales, lustrous turtle carapaces, the shiny-white fangs, claws, and teeth of, variously, the jaguar, puma, ocelot, crocodile, and whale, and the multicolored sheen of innumerable lizards and (sometimes poisonous) frogs and snakes, joined the translucence and color of seashells and pearls in a distinctive set of values which ranged from subsistence to the exercise of political power to the ideational worlds of spirituality and religion. Under the influence of tobacco, for example, shaman-initiates of the Venezuelan Warao experienced strange perceptions of serpents in which touching a snake's shimmering scaly skin could bring spiritual enlightenment (Wilbert 1997: 327; Helms 1993: 212–216)—an event shrouded in billowing clouds of white tobacco smoke that connected the earth and the sky.

The guiding principles that gave shape and meaning to this distinctive Amerindian worldview permeated ideas concerning technology, material culture, and social life, as well as spirituality and ritual. Underwriting these principles, albeit in different cultural ways, appears to have been the significance given to light and its infinitely varied philosophical elaboration.

Composite effigy pendant,
Gran Coclé, 500–850, gold.
GM 5645.29

Detail, frog and snake
effigy pendant,
Gran Coclé, 1150–1520, gold.
GM 5645.217

Frog effigy pendant, Gran Coclé, 1515–1520, gold. GM 5645.27

Double crocodile effigy, Gran Coclé, 500–1520, gold. GM 5645.216

Frog effigy pendant, Gran Coclé, 500–1520, gold. GM 5645.162

Virtually all earthly life is phototropic. When light hits the retina, the visual cortex of the brain is stimulated, and thought and perception are affected (Perkowitz 1996: 31–35). The importance of light on human culture—its aesthetic valuations and cosmological associations—appears to be universal, from Australian aborigines (Morphy 1992), to African tribal societies (Coote 1992: 252–253), and from India to the world of Islam (Bayley 1986: 291–292; and see Betz 1995; Brill 1980; Rivers 1999).

Today, during the recent historical past, and in pre-Columbian times, indigenous American peoples associated light with a mirrored realm inhabited by luminous spirit beings that became visible to shamans and priests in multicolored visions (Goldman 1979: 210; Kensinger 1995: 221; Taussig 1987: 322–323). These visions could be induced by tobacco smoking (Wilbert 1987), ingesting hallucinogens (Furst 1972; Harner 1978), abstinence, and music (Olsen 1975). These associations are sometimes dramatic, as with shaman-initiates of Brazil's Araweté tribe, for whom light-giving tobacco made their skin shine (Viveiros de Castro 1992: 219), and the Akawaio of Guyana, for whom the central concept of spirituality is *akwa* (light, brightness, life) (Sullivan 1988: 423).

Such beliefs existed also in pre-Columbian times. The Aztec had a concept of bright spirituality, where a person's soul was regarded as luminous and "hot." When an Aztec child was sick, his reflection in water was observed; if it was bright, his soul was intact, if dark, it had escaped (López Austin 1988: 204–206, 216). These ideas, manifested in shiny minerals, pervaded Aztec beliefs about the afterlife. Tlalocan, the Aztec rain god's paradise, was a brilliant place lit by divine fire, where human beings appeared as shimmering gems (Burkhart 1992: 89).

Sacred landscapes were nested within the natural world, and the same philosophical ideas were applied to both. The present day Colombian Kogi conceive of snow peaks as gleaming white crystals, prisms of light entered by the dead (Reichel-Dolmatoff 1981: 28). Rainbows and clouds glowed with spiritual energy for indigenous peoples of the Andes (Garcilaso de la Vega 1987 [1609]) and the Caribbean (Stevens-Arroyo 1988: 190–191). For the Zapotec of Mesoamerica, the vital life force, *pèe,* dwelt within all living matter and was manifest in lightning and clouds, the former representing the powerful supernatural Cociyo and the latter the metamorphosed ancestors of the Zapotec people (Marcus and Flannery 1994: 57–58).

What is clear, even from this brief overview, is that the indigenous American world was permeated by a multisensorial appreciation of light, connecting earth, sky, sea, and atmospheric phenomena, suffusing the whole with spirituality and morality, and energizing it with cosmic power. Thus, places, objects, and phenomena that possessed shininess were especially sacred, and rivers, lakes, the sea, mist, rain, and ice were all alive with spiritual essence. Such a world was not understood by 16th-century Europeans and exists only as superstitious folklore fancy in the modern Western mind.

BRILLIANT OBJECTS

The supernatural qualities ascribed to light in indigenous thought were materialized in natural and cultural objects and phenomena. The view that light represented dangerous cosmological energy endowed shiny artifacts with power in the earthly world of politics and religion; in other words, esoteric knowledge was transformed into exoteric action through the manipulation of material culture.

Across the Americas, the most common form such objects took was the mirror: polished iron ore among the Olmec (Carlson 1981); mosaics of pyrite and jade for the Maya (Kidder et al 1946: 126–131) and at Teotihuacan (Taube 1992); anthracite and jet (Burger 1995: 91, 121, 169; Quilter 1991: 404–413) and pyrites and shell (Boone 1996: 181–186) in the Andes; and slate and mica in North America (Ford 1969: 75–76; Saunders 1988: 1–10, fig.1). These were not the vanity mirrors of European tradition, but magical scrying mirrors wielded to give access to and power over the supernatural realm. The mineral nature of such mirrors suggests that minerals themselves were invested with culturally specific philosophical qualities. In Mesoamerica, the Spanish priest and chronicler Bernardino de Sahagún (1950–78, bk. 11: 223, 229–230) refers to the translucent preciousness, iridescent colors, and light-giving qualities of different minerals, many of which were recorded in the Aztec Codex Mendoza (Berdan 1992: 310–312). Still today, modern Nahuatl Indians see the sky as a living (crystal) mirror shining with the sparkle of the sun and stars (Sandstrom 1991: 238; Dow 1986: 108–110). In the Andes, crystals adorned Inka temples (Mester 1990: 206, 216) and were associated with pearls and rain. For the Amazonian Desana, a crystal's hexagonal shape represents cosmic order. It is related to energy and fertility and considered as concentrated human semen (Reichel-Dolmatoff 1975: 102; 1981: 23). Among the Chippewa-Ojibwa and Seneca Iroquois of North America, shamans used quartz crystals for curing and divination (Hamell 1992).

Other shiny minerals shared these qualities. Mica symbolized cosmic forces. It was associated with high status burials and buildings in Mesoamerica, from the earliest times in Oaxaca around 1150 BCE (Marcus and Flannery 1996: 101–103) to Teotihuacan (Armillas 1944: 123–124) and the Aztec civilization (Sahagún 1950–78, bk. 11: 235). Polished greenstones (especially jadeite and nephrite) were particularly valued. The Aztecs, for example, saw them as representing fertility and glistening preciousness (Sahagún 1950–78, bk. 11: 221–222).

Artifacts made from or incorporating translucent shells and pearls also shared in the power of brilliance. Shells were associated with reflective watery surfaces and the fertility aspects of the sea, rivers, and lakes (Cordy-Collins 1990; Kolb 1987). For the Aztecs, the shell of *Spondylus* princeps resembled crystal and was used to make shamans' divinatory bowls (Berdan and Anawalt 1997: 84). In the Andes, shells were commonly offered at mountain water sources (Murra 1975: 257), and in Amazonia, shell jewelry was (and remains) highly esteemed (Medina 1934: 411–412). In North America, shell beads were symbols of saliva and semen (Hall n.d.; Ceci 1988), and, when made into wampum (cylindrical shell beads), represented light and spiritual well-being (Hamell n.d.; 1995). Pearls similarly possessed light-giving and spiritual qualities (Mester 1990: 198; Saunders 1999: 247–249).

Wampum, 18th century, shell, cotton string. GM 9026.815

Conch shell gorget from Venado Beach, Panama.

Dumbarton Oaks, Pre-Columbian Collection, Washington, D.C.

Amazonian feather headdress.

Museum of the Red River, Idabel, Oklahoma

Burnished pottery and ceramics containing sparkling inclusions dazzled with their shininess (Arnold 1993: 113; Medina 1934: 201), as did polished black wood in the Caribbean and shimmering featherwork in lowland Amazonia (Howard 1991; Kensinger 1991). All these objects resonated with the positive cultural values and associations of light and reached out into the wider worlds of metaphor and ritual activity. Whatever the medium, there appears to have been an underlying structure of meaning whereby sacred power was manifested in an object's reflective and chromatic qualities. The Amerindian world was a brilliant place.

THE POWER OF BRILLIANCE

Pre-Columbian peoples saw the cosmic powers of light in three-dimensional objects and places. Philosophies of life and death were everywhere apparent—reified in material culture, mountains, rivers, and clouds. In this universe, making shiny objects was an act of transformative creation. Such activities trapped and converted the fertilizing energy of light into brilliant solid forms through technologies that drew their efficacy from a synergy of myth, ritual knowledge, and individual skill (Saunders 1999: 246).

Pregnant women among Muisca people of Colombia, for example, offered emeralds and figurines made of gold-and-copper alloy to their Rainbow god to ensure successful childbirth. In this way, golden artifacts and gemstones interacted with an iridescent atmospheric phenomenon (produced by sunlight shining through moisture) to enhance human fertility. This interleaving of natural and cultural worlds demonstrates how deeply held philosophical ideas imbued artifacts with sacred power. Golden objects were not artworks (as in our Western view), but spiritually active entities that maintained the world. Shiny objects are revealed as the ultimate symbols of power and influence.

The power of light, whether appearing in natural phenomena or "captured" in shiny objects, was understood to be a characteristic of, and controlled by, ancestors, spirits, deities, and immanent forces. Only those with sacred knowledge and privileged access to these omnipotent but ambivalent supernatural beings were able to wield this power on earth. As supernatural beings and supernatural phenomenon were enveloped in light, so those who exercised power on earth were symbolically and physically surrounded by shininess and color. Thus, shamans, priests, sorcerers and dynastic rulers—who manipulated glittering power objects such as mirrors, metals, jade, and gemstones—were the spiritual and political power-brokers of the pre-Columbian world.

These high status individuals literally and figuratively dazzled their audiences in brilliant costumes, paraphernalia, and rituals. They fascinated, compelled, entrapped, and delighted spectators with their colorful and magically brilliant objects (Gell 1998: 23) that were understood to be cosmic energy in solid form. It was hardly surprising

that during Sir Walter Raleigh's first expedition to the Orinoco, a local chieftain forbade his people to trade in gold because it undermined his authority (Whitehead 1997: 77). Among the Taíno of Hispañola, some chiefs had names such as Caonabo and Anacaona, which incorporated their word for gold, *caona.*

GOLDEN SYMBOLS

Gold, golden objects, and alloys were not the sole possessors of magic power in pre-Columbian America, though they obsessed later European explorers and conquistadors. Shiny metals derived their indigenous values from the deeper and already well established philosophy concerning light and color. Gold and gold alloys were received into this ancient world of phenomenological experience. Metals, nevertheless, did possess qualities that enhanced the magical power of all brilliant objects. This was partly due to the complex technologies that had to be mastered in order to produce finished objects—technologies that employed the "magical" transforming power of fire. Practical and ritual processes combined to produce highly reflective metals that were able to gather and dispense more light than minerals, shells, or feathers. The philosophical and sensorial effect of metals was, at least initially, probably the more intense because, unlike their shiny predecessors, these new conveyors of brilliance did not exist in nature—they could only be created by human beings (with supernatural help).

For pre-Columbian peoples, metals were sensorial stimulants used for decoration and adornment, and to symbolize status and power by displaying the connections between high-status individuals and cosmic beings and forces. The Amazonian Desana believe that the true value of metal adornments lies in the symbolic associations of their color, shape, and odor and their ability to modify colorful hallucinogenic visions (Reichel-Dolmatoff 1981: 22–23; Hosler 1994: 241–243). Copper ear pendants were associated with Desana men's virility, and the smell of the metal related to sex and fertility (Reichel-Dolmatoff 1981: 22).

Non-western ideas about gold and other metals were a feature of this pre-Columbian fusion of metalworking and cosmology. Indigenous metalsmiths often preferred the copper-gold alloy—known as tumbaga in South America and guanín in the Caribbean—to pure gold, possibly in part because of its color [see Quilter, this volume]. Cosmic energy, which resided in the gold and gilded copper ornaments of the Colombian Kogi, could be recharged by exposure to sunlight (Reichel-Dolmatoff 1981: 26). Metals, it seems, partook of ancient beliefs concerning sacred brilliance, but were recognized to be the most effective conveyors of light. As such, they often complemented rather than supplanted the translucent shimmer of crystals, jade, and feathers, as well as the brilliance of shamanic visions, lightning, and snowfields.

Embossed disk, Gran Coclé, 500–1520, gold. GM 5645.59

Auri legendi ratio in rivis è montibus Apalatcy decurrentibus. XLI.

ROCVL ab eo loco, in quo nostra arx extructa fuit, magni sunt montes, Indorum lingua, Apalatcy cognominati, è quibus, ut ex topographica charta videre licet, oriuntur tres magni rivi, provolventes arenam, cui multum auri, argenti & æris admixtum est. Eam ob causam illius regionis incolæ fossas in rivis faciunt, ut provoluta ab aqua arena, in eas propter gravitatem cadat: diligenter inde eductã in certum locum deferunt, & aliquanto post tempore, denuo fossis arenam quæ incidit exhaurientes, colligunt, & cymbis impositam per ingens flumen devehunt, à nostris Maii nomine insignitum, quod in mare sese exonerat. Opes quæ inde proveniunt, nunc Hispani in suum usum convertere norunt.

Pre-Columbian metalworking employed specialized techniques, but these were still governed by the beliefs and metaphysical principles that connected society to cosmological forces. Yet, even as metalworking developed in pre-Columbian America, its existence modified the social relationships of those societies which practiced it (Pfaffenberger 1992: 500). Perhaps not least of these changes was, as far as we can tell, that metalworking, unlike pottery or textile production, was a uniquely male preserve. Even the collection of gold-bearing ores was carried out without the presence of women (Cooke et al 2003:110).

It may be that the perceived dangers of preparing for and working such cosmically-charged material was regarded as too risky for women—not in a physical sense, but in terms of how indigenous peoples saw their sexuality related to cosmological beliefs about the fundamental processes of conception and birth. The creation of metals (by fire, technology, and "magical knowledge") in crucibles and furnaces can be likened to new human life growing (i.e. being transformed) in the womb (Falchetti 2003: 356), an analogy which was certainly drawn with the production and use of pottery, where it is used to cook food and was likened to the womb which "cooked" the fetus into a human being.

The Colombian Desana offer a valuable insight into the idea that creating metalwork was regarded as transforming raw materials. While women may not have been involved with metalworking, they were implicated in the value system which underwrote such activities. The relationship between metalsmithing and the reworking of cosmic energies was particularly gendered when it came to the alloying of copper with gold. The redness of copper and its smell were both associated by the Desana with female fertility. Red was associated with blood and transformation and allied to the embryonic transformation of the unborn child in the womb (Falchetti 2003: 350). The alloying of metals represented in material culture the blending of male and female properties and thereby established cosmic balance (ibid.: 359).

Regulating these powerful life forces is the responsibility of those individuals who bring order to matter through their knowledge of ritual. The mystery surrounding metalworking as transformative creation is analogous to the special knowledge and activities of those engaged in curing, divination, and "creating socialized people" through directing rites of passage. Successful metalworking is revealed as a synergistic mix of technological expertise, ceremonial activity, and ritual knowledge.

Facing: Amerindians Panning for Alluvial Gold, Theodore de Bry, *The American Indians Depicted in a Collection of Plates Engraved by Theodore de Bry in the Years 1590–1602.* GM 2975.1585

Greenstone figurine from Panama/Costa Rica. Dumbarton Oaks, Image Collections and Fieldwork Archives, Washington, D.C.

GOLDEN PANAMA

Ancient Panama, located between Mesoamerica and South America, was inhabited by indigenous peoples who shared the worldview of their northern and southern neighbors, including the aesthetic of brilliance objectified in shiny material culture. Yet, pre-Columbian Panamanian societies were also on the "metal route," as metalworking began in the Andes around 1800 BCE and moved northwards to arrive in Mesoamerica by around 900.

Before goldworking arrived in the region of Panama and Costa Rica, greenstones (especially jadeite and nephrite) were the favored carriers of sacred brilliance. Throughout Amazonia, greenstones were highly valued and widely traded, as they were also in the Caribbean (Boomert 1987; Oliver 1989: 216-217). In Mesoamerica, similarly, and from at least Olmec times (ca. 1250 BCE–400 BCE), jade was a sacred material. It remained so until the Aztec era, at which time greenstones symbolized fertility and preciousness, signaling their presence at dawn by emitting smoke and imparting greenness to the flora in their vicinity (Sahagún 1950–78, bk. 11: 221–222).

There is evidence that during Aztec times gold or gold alloys were beginning to be valued as highly as greenstones, though they had not yet displaced them (see Baquedano 2005). Five hundred years after the arrival of metalworking in Mesoamerica, gold had not replaced jade as the favored conveyor of brilliance, though it had begun to share at least some of the underlying symbolic qualities of shininess.

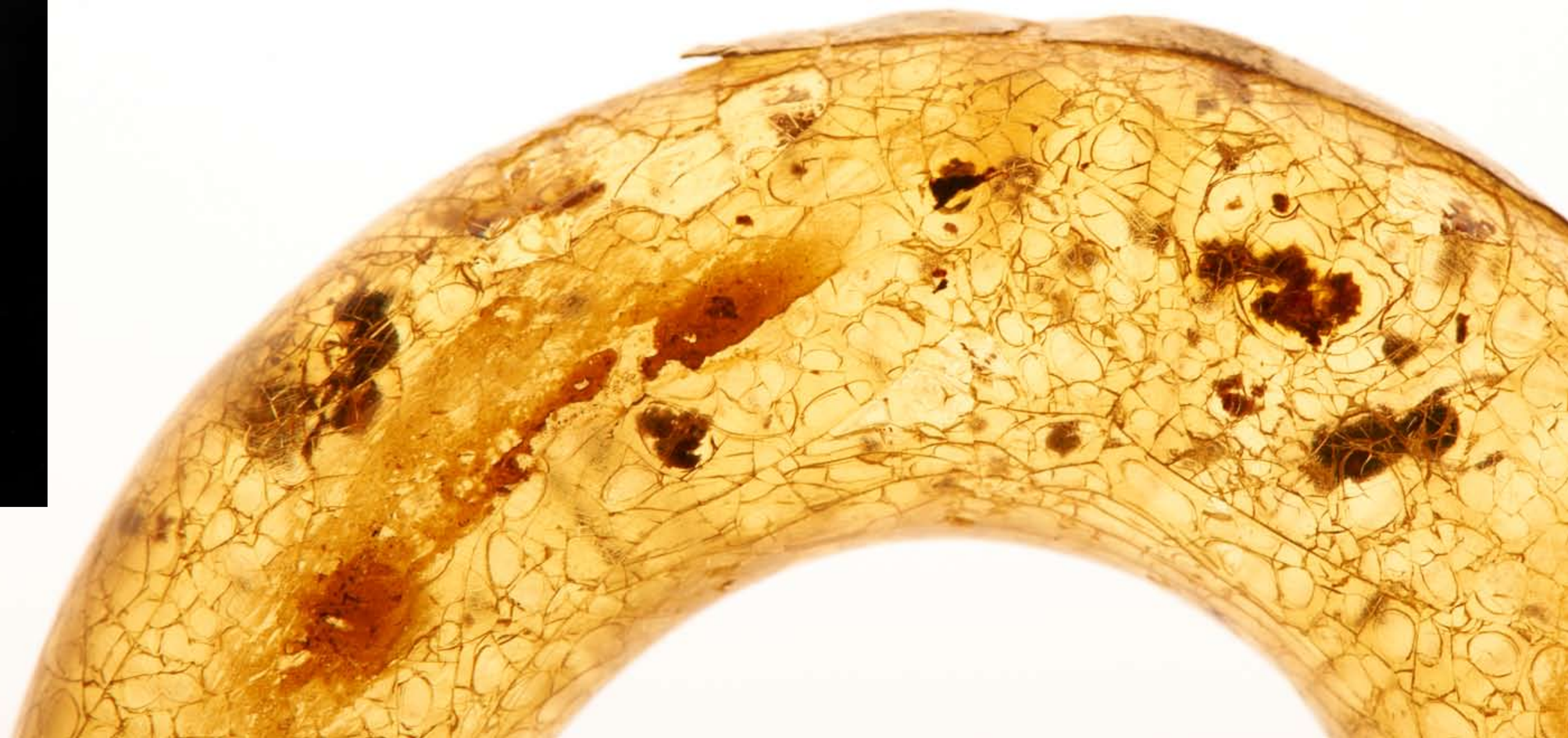
Nose ornament, Gran Coclé, 700–1000, gold, greenstone. GM 8445.2104

Bracelet, Gran Coclé, 500–1520, gold, greenstone. GM 5645.233

In Costa Rica, the jade axe tradition and examples of reworked Maya jade celts can be seen as pre-Columbian Costa Rican societies' understanding and reworking of such power objects for their own societies (Graham 1993: 22–26; Bruhns 1996: 290; Lange 1992: 118). Equally possible is that Mesoamerican reworked and "imitated" jade artifacts were also sophisticated brilliant objects from distant realms, at a time when metallurgy was little known or unknown further north and had not yet displaced other shiny objects in Lower Central America.

Facing: Tooth pendant, mammalian (possibly feline) canine tooth encased in gold. Gran Coclé, 700–1000, gold, bone. GM 5645.107

Right and below: A composite gold and amber nose ring (two views), Gran Coclé, 700–1000, gold, resin. GM 5645.105

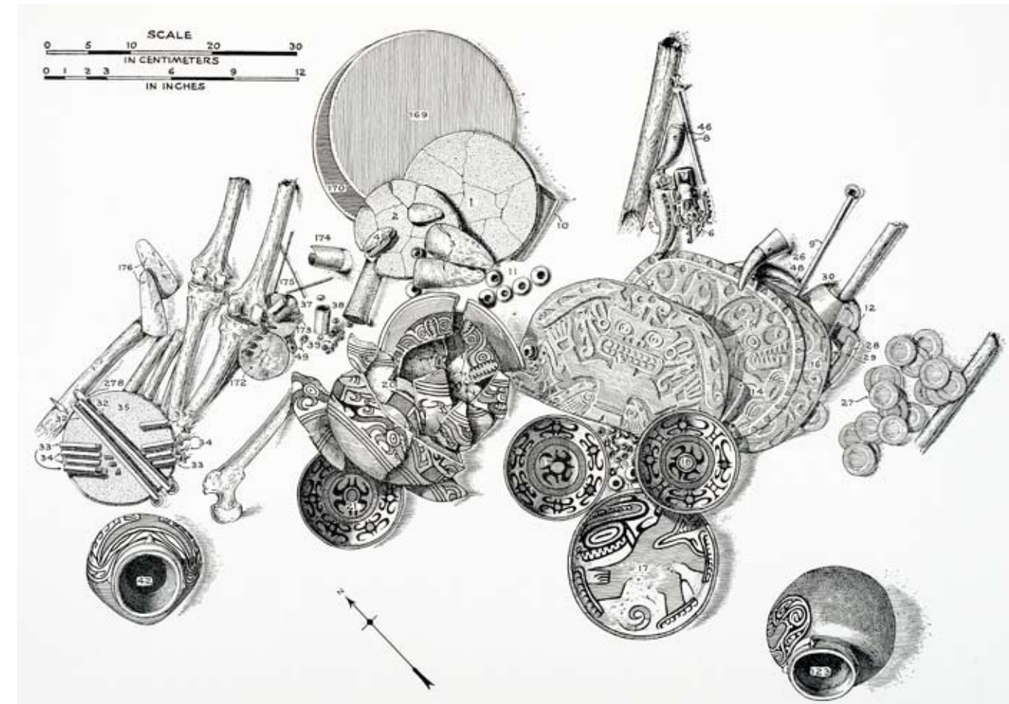


Circular golden disks such as this were presumably worn as chest ornaments. Embossed disk, Gran Coclé, 500–1520, gold. GM 5645.57

Evidence from Costa Rica between 300 BCE and 500 suggests that carved and polished jade artifacts served as symbols of rank alongside gold objects (Snarskis 1992: 144-145). In Panama at this time, many gold objects incorporate other shiny or colorful materials such as greenstone, shell, amber, and ivory. At the sites of Cerro Juan Díaz and Playa Venado, blindingly reflective gold plaques, or disks, were buried alongside shell beads (Cooke et al. 1998: 139; Cooke et al. 2000: 161-1 Cooke et al. 2003: 63). Such discoveries demonstrate that when gold objects first appeared in Panama and adjacent areas, they were initially valued equally with long-appreciated shiny minerals and shells.

These earliest of gold items were likely imports from further south (in modern day Colombia), and it was not until somewhere between 500 and 750 that goldworking *in situ* appeared in the region (Cooke et al 2003:133; Snarskis 2003). The first locally produced gold objects, perhaps unsurprisingly, appear to have imitated items previously made from shells [Quilter, this volume]. The peerless shininess of gold and gold alloys could not be denied, however. Sometime after 500, such items overtook polished minerals and shells as the favored carrier of sacred brilliance and its connotations of cosmically-sanctioned high social status and political power.

In Panama, there is little doubt that the arrival of goldsmithing had "a profound ideological impact on local society…" and that "metalwork artefacts [at the famous burial site of] Sitio Conte epitomize what seems to be a rather sudden change in attitudes towards wealth subsequent to the introduction of the technology: a concern for extravagant display on both the living and the dead" (Cooke and Ranere 1992: 286). In neighboring Costa Rica, this same transition has also been observed (Snarskis 1998: 90).



Plan of deposit XVI, grave 26, Sitio Conte, Panama. The objects shown include: stone mirror backs (1, 2, 35), five stone celts (3, 176), pottery incense burners (169, 170), gold plaques (13, 14, 15, 16), gold arm bands (10, 28, 29, 30), 25 gold disks (27), gilded disks (172, 173), gold necklaces (11, 39, 49), four gold ear rods (8, 9, 32), four gold cylinders (37, 38), two gold and stone ear rods (33), tips for two wooden ear rods (34), three gold chisels (175), emerald (7) and gold setting (6), whale tooth pendant (46), carved whale tooth pendant (5), pair of whale tooth pendants set in gold (26, 48), chipped stone blades (4), and pottery vessels (17, 18, 19, 20, 21, 42, 123).

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In Panama and Costa Rica, the superior reflectivity of metal adornments over those of jade and other minerals might suggest an increase in the elite's control of bright cosmic power and a furthering of social distance between rulers and ruled. While gold was almost certainly worn in multimedia displays of minerals, textiles, feathers, feline canines, and shell, the net result of the new metalworking technology was that in ritual and everyday events, chiefs, priests, and shamans would have appeared more blindingly brilliant than before.

The intimate symbolic relationship between shininess, metal, and high social status is seen to great effect at Sitio Conte, whose extraordinary wealth of golden objects are dated to between 750 and 950 (Cooke at al 2003:117). Given that metalworking was moving north from South America, it is not surprising that the earliest Panamanian gold objects derived from adjacent Colombia and the various regional metalworking styles that are found there (Bray 1992: 34).

The goldworking style called by archaeologists the International Group (400–900) yields particular insights, as, during this time, the region that stretched from central Colombia to northern Costa Rica is regarded as having been one metallurgical province. Despite local variations (Bray 1992: 35), the peoples who lived here shared a single goldworking tradition, making animal figurines, human figures, and bells, alongside circular pendants.

Headband, Gran Coclé, 500–1520, gold. GM 5645.108

Bracelet, Gran Coclé, 500–1520, gold. GM 5645.190

It may be that such golden images were regarded in some way as spiritually, culturally, and politically neutral—hence their acceptance over a wide area and among different groups. Equally possible is that as a new technology these highly-polished shiny objects were accepted as novel—perhaps intensely powerful—conveyors of sacred light (and all its symbolic connotations). Their superior reflectivity was appreciated due to the prior tradition of valuing shininess. Animal and human shapes were commonplace (in life and art), but their appearance in a uniquely brilliant material whose magical production was presumably surrounded by shamanic mystique would have made a powerful psychological impact around 500.

In Panama, the aesthetic of brilliance found cultural expression also in a shiny and colorful pottery tradition (Linares 1977; Helms 1993). In her study of Panamanian polychrome ceramics, Mary Helms argues that we should look "beyond style to consider underlying meaning and symbolism" (Helms 1993: 210) by exploring possible associations of brilliance and color in animals, their individual roles in mythology, and the way they are represented in iridescent pottery. Helms makes a prescient remark in her conclusion—that we might consider that "metallurgy and polychrome ceramics expressed pure light and energy and the colourful constituent components of that light and energy, respectively" (ibid.: 245).

Human effigy pendant, Gran Coclé, 500–1520, gold. GM 5645.73

In ancient Panama, there appears to have been an abundance of different kinds of shiny media—from pearls and shells, to minerals, pottery, and gold and gold alloy objects—that, despite their distinct origins, shared the quality of brilliance. It was this quality which made their acquisition, production, and display such a widespread cultural choice for representing social status, inasmuch as by "wearing light," chiefs, priests, and shamans demonstrated their access to and control of supernatural cosmic power.

CONCLUSION

The idea that Panamanian gold artifacts were powerful cosmological implements rather than simply shiny jewelry is intellectually challenging. The notions behind the aesthetic of brilliance argue that an ancient and widespread belief held all shiny things (and phenomena) to be charged with the spiritual power of ancestors and the supernatural. In this view, when metallurgy is invented and moves from South America northward through Panama, its shiny new golden objects are received into a preexisting framework where reflected light and brilliant colors are already understood to be a manifestation of the cosmic energy that regulates the universe and all life within it. This view of the world was tenacious. It endured up to the Spanish conquest of the 15th and 16th centuries, and even to the 18th, when gold birds were still being cast in southern Costa Rica (Cooke and Bray 1985:1). Such activities were commented upon in many European accounts of initial contact with the region's indigenous peoples. These accounts tell of the gold-wearing habits of local elites and their seemingly obsessive accumulation of such objects.

Ferdinand Columbus observed the fashion amongst native leaders on the Caribbean coast for wearing "gold mirrors and eagles" (quoted in Cooke and Ranere 1992: 286), and Gaspar de Espinosa tells of one chief, Caubaco, who wore a thousand castellanos of gold when raiding other tribes. Fernández de Oviedo reported that "it was the custom in those parts for the chiefs and important men to bring to battle some gold jewel on their chests or head or arms in order to be known to their own men and also by their enemies" (Oviedo 1853: 118, quoted in Cooke this volume).

Such reports have been interpreted as war leaders wearing golden costumes and paraphernalia so that they could be seen on the battlefield for tactical advantage. Yet colorful banners and perhaps feathered headdresses could equally well serve this straightforward military purpose. Given the cosmic values ascribed to shiny objects, and to gold in particular, it is equally likely that by being sheathed in gold, war chiefs were advertising both their physical access to and ritual control of supernatural energy and brilliance—in other words, spiritual power. The more gold they wore, the more light they reflected, and the more powerful they appeared to be. On the field of battle, indigenous philosophies concerning the spiritualized mechanisms of life and death had real world consequences.

Gaspar de Espinosa was also an eyewitness to a startling manifestation of the indigenous Panamanian desire for gold and gold alloys. By chance, in 1519, he interrupted the burial rituals of Chief Parita, and left the following account of what he saw.

> … on his head a great basin of gold … and at his throat four or five necklaces … and on his arms casings of gold like cylindrical tubes … and on his breast and shoulders many plates and other pieces of like coins, and a belt of gold, all surrounded with gold bells, and on his legs also the same golden armor … in such a way that the body of the cacique [chief] was armoured, he seemed to be in a coat of mail or suit of armor (Espinosa quoted in Linares 1977: 76).

This late example was but a Panamanian version of the wider pan-American attraction to brilliant objects that were seen as materializations of otherwise untouchable cosmic power. A chief from Darién told the Spanish that his golden ornaments came "from the sky" (Helms 1979: 87)—exactly the same story given to the Spanish by the Taíno of the Greater Antilles (Chanca 1847: 64). In other words, shiny objects were pieces of the cosmos fallen to earth.

Panama, and the adjacent areas of Costa Rica and Colombia, offer deep insights into the ways that indigenous Amerindian societies conceived of the universe and their place within it through the socially and symbolically transformational medium of material culture. Modern interdisciplinary archaeology allows us to track and interpret the movement of metalworking northwards from its South American place of origin, and its multi-layered effects on societies receiving it for the first time. Along parts of Panama's Pacific coast, for example, a change in ceramic traditions after 700 may have been associated with reconfigured social and economic interactions as gold objects replaced shell jewelry as the focus of exchange networks (Cooke 1998, and this volume).

In a universe governed in Amerindian eyes by the symbolic ebb and flow of cosmic forces, the power of an object (and the technological processes required to produce, or otherwise obtain it) was framed by and enacted through sacred ritual knowledge. Such knowledge was not simply esoteric but actively political; it embodied natural and moral philosophies and bestowed social meaning on objects. Reworking the dangerous but infinitely powerful stuff of creation into wearable items was a manifesto of spiritual and political power—and golden objects its peerless icons.

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